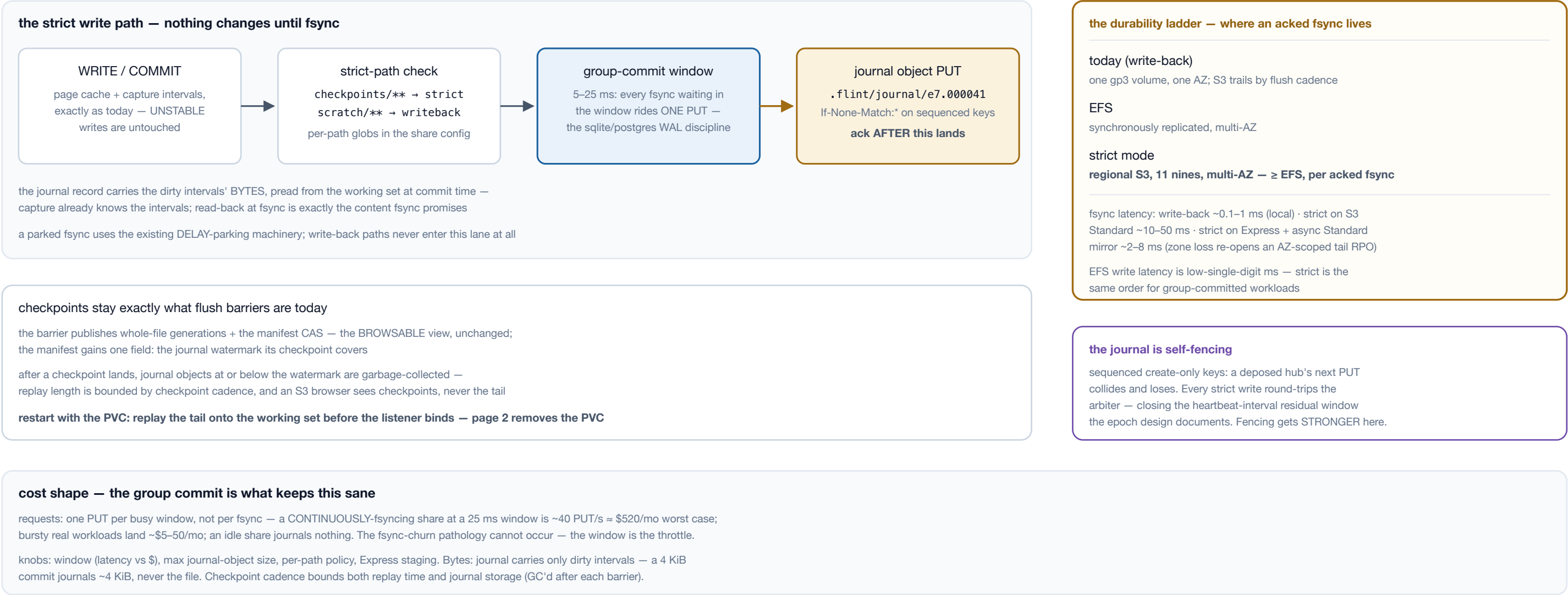


A per-share, per-path upgrade to the hub: an acked fsync is durable in regional S3 — a stronger guarantee than EFS's replication — bought with a group-committed journal beside the unchanged browsable checkpoints. The UI, the bucket format, and write-back paths are untouched; only flagged paths pay the latency.



Once the journal exists, S3 is the authoritative log and the hub can be rebuilt anywhere from the bucket alone: checkpoint + journal replay + protocol grace. The force-detach wait, the AZ pinning, and the kubelet staging-wedge class all hang off the PVC being load-bearing — and stop existing when it isn't.

the failover ladder

today — the PVC is load-bearing

pod crash ~13 s · node crash: force-detach up to ~6 min, AZ-pinned, F29 staging wedges need a manual arm

journal rebuild — successor starts ANYWHERE

no volume reattach, no AZ pinning, no kubelet staging state — checkpoint pull + tail replay + grace: tens of seconds, bounded by checkpoint cadence and grace, not by EBS

warm standby (v2) — a second pod tails the journal continuously

takeover ≈ fence + grace; the epoch CAS + journal-seq collision already arbitrate who serves

which failure-track rows this improves — and which it does not touch

F4 power loss / F1 crash storms — acked strict writes get a second, independent durability witness

F8 disk-full, F13 quarantine fallback — recovery no longer trusts the disk that just failed

F10 node death — the ~6-min force-detach row is deleted; F29's kubelet staging-wedge class cannot occur

split-brain — the epoch heartbeat's documented residual window closes (journal seq is the fence)

NOT touched: the NFS protocol surface (conformance, quotas, deadlines — the B1–B10 work stands on its own and is a PREREQUISITE here, not a beneficiary); coherence (the hub is still the one authority); the CL load track (concurrency behavior is orthogonal to where durability lives)

strict mode is the milestone AFTER the hardening drills, standing on them — not a replacement for them

the rebuild, step by step — no PVC anywhere

- 1 schedule the pod on any node, any AZ — working set on an empty volume (now a cache)
- 2 import the last checkpoint: manifest → dirs, symlinks, evicted stubs (the existing DR verb, unchanged)
- 3 replay the journal tail — touched files hydrate, journal bytes land at their offsets; idempotent
- 4 fence: first journal PUT at seq N+1 — the old incarnation's straggler collides and loses
- 5 fresh state.db + FULL protocol grace: clients reclaim locks and opens — the B1/B2 reclaim fixes and the B3 verifier are exactly what make this client-transparent (retransmitted WRITES land, locks return)
- 6 serve — clients saw a stall and a reboot, never an error; acked strict writes all present

recovery ground truth is the BUCKET, not local disk — the state.db corruption and disk-full legs of the failure track get an independent witness they have never had

deliberately NOT in the journal

sessions, client records, open/lock state — rebuilt by the PROTOCOL (grace + reclaim), not by replay; journaling them would duplicate machinery NFSv4.1 already specifies and this repo already hardened

write-back paths — their contract is unchanged; the journal covers strict paths only, so the tail stays small and replay stays fast

one engine, three front ends — not lite-hub-specific

the journal is tier:: functionality: the same lane is the strict mode for lean and FUSE sidecars (their fsync ⇒ S3 story on page 2 of the consistency comparison), and the lean gateway is its natural group-commit point. Build once, every front end inherits the durability knob.

out of scope: the full-pNFS/DS posture (the tier is scoped lite-only by design review A1)

what it takes to build

model first (~1 wk): journal-seq fence × epoch × checkpoint watermark — the FlintTierSession discipline. Phase A, strict journal (~4–6 wk): journal writer + group commit ~1.5k lines · fsync parking hooks ~500 · policy ~150 · checkpoint watermark + GC ~300 · replay-on-restart ~400 · drills. Phase B, rebuildable hub (~4 wk): no-PVC rebuild path ~1k · fence rework · operator failover + cache-mode chart · drills. Standby v2: ~2 wk. Ships per-share, opt-in.

The build, honestly costed: a formal-model week, then Phase A (~4–6 engineer-weeks) delivers the EFS-exceeding durability guarantee as an opt-in per-share knob with the PVC still present, and Phase B (~4 weeks) makes the PVC a cache and collapses the availability gap; a warm standby is a further ~2 weeks when wanted. Almost every hard prerequisite already exists: capture's dirty intervals, the guarded store trait, the DELAY-parking machinery, manifest/import as the checkpoint verb, the epoch CAS — and, critically, the hardening campaign's own fixes: **B1/B2's reclaim correctness and B3's incarnation verifier are what make a rebuilt hub client-transparent**, so this design stands on the hardening story rather than competing with it. The division of labor is clean: the CL/F drills earn the right to ship the hub at all; strict mode is the first post-GA milestone, converting the two failure-track rows that drills can only measure (PVC loss, node-death stalls) into rows that no longer exist. And because the journal is tier-engine functionality, the same lane is the strict mode for the lean and FUSE sidecars later — one build, three front ends, with the fencing getting stronger as a side effect.